



CSE422: ARTIFICIAL INTELLIGENCE

Lab Project Report

**Topic: Depression Prediction Using Machine Learning**

Project Type: Classification (Supervised & Unsupervised Learning)

Group No: 12

Section: 03

**Group Members:**

| Name                 | ID       |
|----------------------|----------|
| Mahedi Mostafa Rifat | 22101739 |
| Shad Nur Mim Bidhu   | 20101574 |

Department of Computer Science and Engineering

BRAC University

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### **1. Introduction**

In recent years, mental health issues among university students have become a critical concern. Academic pressure, financial instability, and social expectations often contribute to high levels of anxiety and depression. This project aims to build an automated Artificial Intelligence system that can predict a student's depression level based on their academic and lifestyle factors.

The motivation behind this project is to explore how Artificial Intelligence can assist in early diagnosis. By analyzing patterns in students' sleep, academic performance, and stress levels, we aim to develop a model that can classify students into different risk categories (e.g., Severe, Moderate, or No Depression), potentially helping counselors intervene earlier.

### **2. Dataset Description**

#### **Dataset Overview**

- Total Data Points: 1,977 student records.
- Features: 37 features (e.g., Academic Year, CGPA, Sleep Habits, Anxiety Scales).
- Target Variable: Depression Label (Classification Problem).
- Feature Types:
  - Categorical: Gender, Academic Year, Sleep Status.
  - Quantitative: Age, CGPA, Anxiety Score.

## Why Classification?

The target variable (Depression Label) represents depression severity levels categorized as

- 0: Mild Depression
- 1: (not specified in output)
- 2: Moderate Depression
- 3: Moderately Severe Depression
- 4: (additional categories)

Since we are predicting discrete categories rather than continuous values, this is a multi-class classification problem.

## Feature Types

The dataset contains a mix of quantitative and categorical features:

Demographic Features:

- Age (Categorical: age ranges like 18-22, 23-27, etc.)
- Gender (Categorical: Male/Female)
- Academic Year (Categorical: First Year, Second Year, etc.)
- Current CGPA (Categorical: CGPA ranges)
- Scholarship Status (Categorical: Yes/No)

Anxiety Indicators (7 features):

Questions measuring frequency of anxiety symptoms (rated 0-3):

1. Feeling nervous, anxious or on edge due to academic pressure
2. Unable to stop worrying about academic affairs
3. Trouble relaxing due to academic pressure
4. Easily annoyed or irritated because of academic pressure
5. Worried too much about academic affairs
6. Feeling restless and hard to sit still
7. Feeling afraid as if something awful might happen

Stress Indicators (10 features):

Questions measuring academic stress levels (rated 0-4):

1. Feeling upset due to academic affairs
2. Unable to control important academic things
3. Feeling nervous and stressed
4. Unable to cope with mandatory academic activities
5. Confidence in handling academic problems
6. Things going your way academically
7. Ability to control irritations
8. Academic performance on top
9. Anger due to poor performance
10. Academic difficulties piling up

Depression Indicators (9 features):

Questions measuring depression symptoms (rated 0-3):

1. Little interest or pleasure in doing things
2. Feeling down, depressed or hopeless
3. Trouble falling/staying asleep or sleeping too much
4. Feeling tired or having little energy
5. Poor appetite or overeating
6. Feeling bad about yourself or like a failure
7. Trouble concentrating on things
8. Moving/speaking slowly or being restless
9. Thoughts of being better off dead or hurting yourself

Aggregate Scores:

- Anxiety Value (sum of anxiety indicators)
- Anxiety Label (categorical anxiety level)
- Stress... (stress-related aggregates)
- Depression Value (sum of depression indicators)
- Depression Label (target variable)

## Correlation Analysis

We performed a correlation test to understand the relationship between different academic pressures and mental health scores.

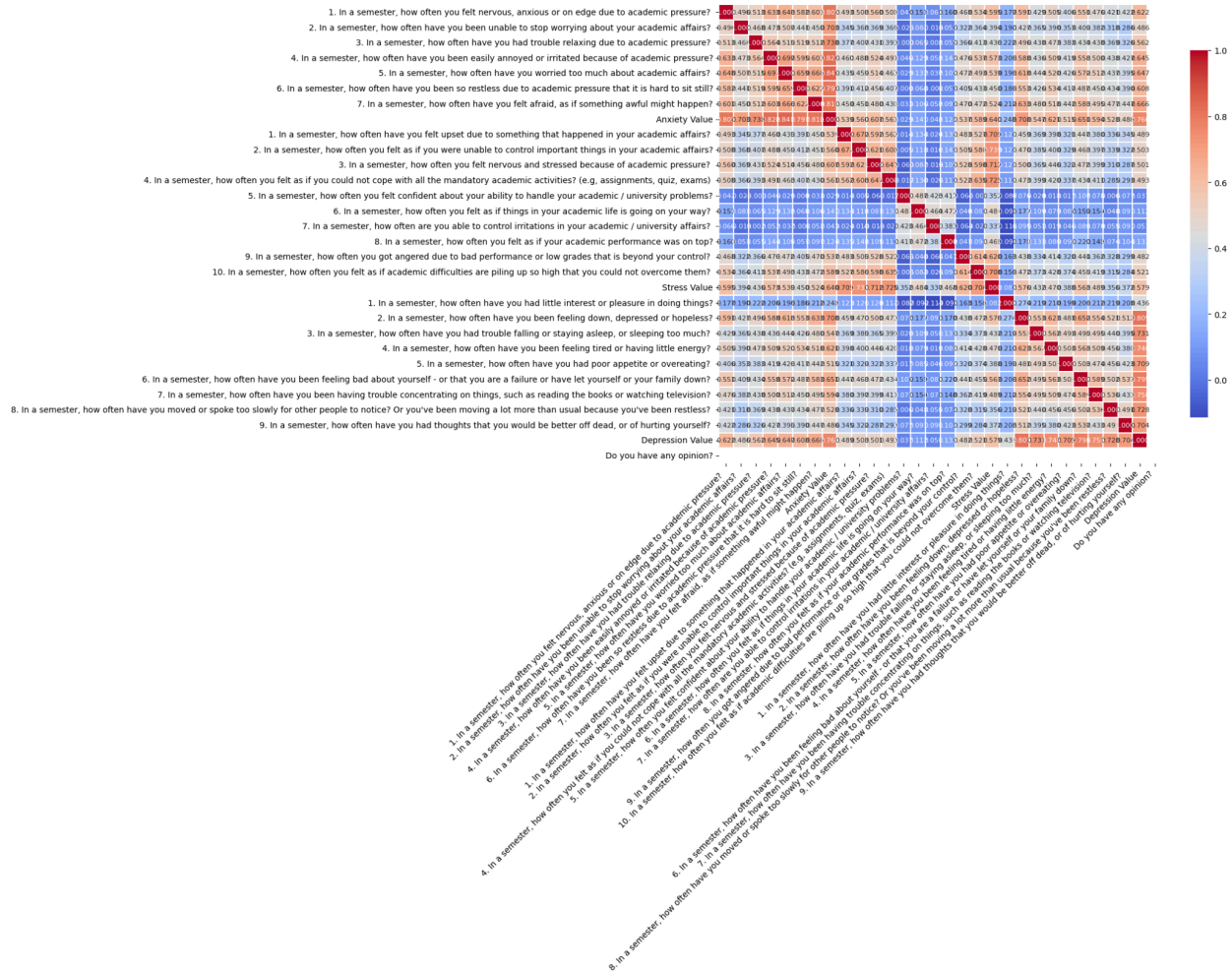


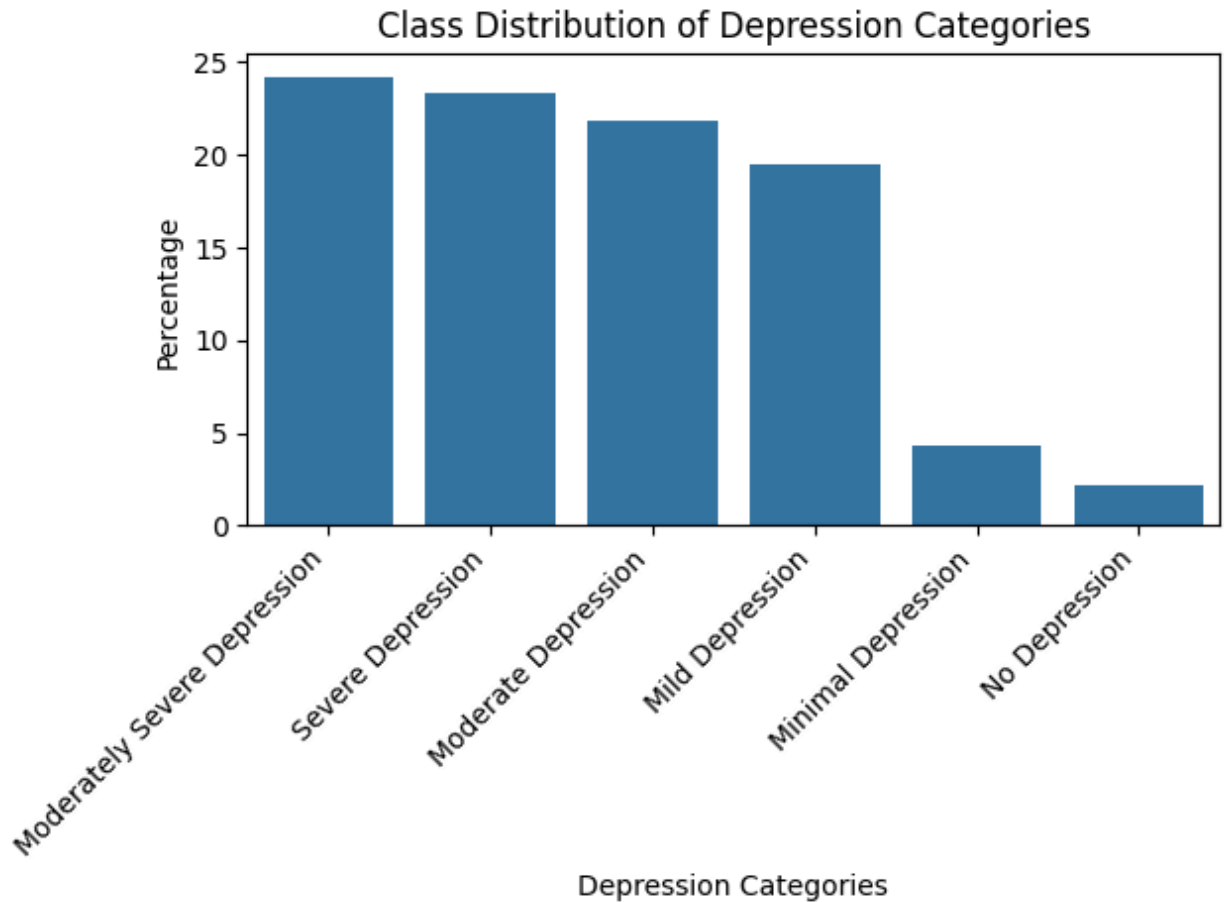
Figure 1: Correlation Heatmap showing relationships between numerical features.

## Observation:

The heatmap reveals a strong positive correlation between "Anxiety Value", "Stress Value", and "Depression Value". This suggests that students with high academic stress are significantly more likely to suffer from depression.

## Imbalanced Dataset Analysis

We analyzed the distribution of our target variable (Depression Label) to check for dataset imbalance.



*Figure 2: Distribution of Depression Labels across the dataset.*

### Observation:

The dataset is not perfectly balanced; some categories appear more frequently than others. To address this, we used Stratified Splitting during the training phase to ensure our model learns to recognize all classes equally.

## Encoding Categorical Variables

*Do we need to encode categorical variables?*

Yes, encoding is required because machine learning algorithms work with numerical data. The categorical variables (Age, Gender, Academic Year, CGPA, Scholarship Status, etc.) contain string/text values that need to be converted to a numerical format.

**Encoding Method Used:** Label Encoding was applied to convert each categorical value to a unique integer using sklearn's LabelEncoder.

## 3. Dataset Pre-processing

### Problem 1: Missing Values

Fault Identified:

Upon inspection, the dataset contained missing values in columns like "Gender" and "Academic Year". Additionally, to simulate real-world imperfections as per project requirements, we introduced random null values (5% of data).

Solution Applied:

Instead of dropping rows (which would lose valuable data), we applied Imputation:

- Numerical columns: Filled with the Mean (average).
- Categorical columns: Filled with the Mode (most frequent value).

This ensured we preserved all 1,977 records for training.

### Problem 2: Categorical Values

Fault Identified:

Categorical columns (Gender, Academic Year, Sleep Status) contained string values (e.g., "Male", "Fourth Year") that cannot be processed by ML algorithms like Neural Networks.

Solution Applied:

Label Encoding was applied to convert these text labels into unique numerical integers.

```

1 #Encode Data (Label Encoding)
2
3 le = LabelEncoder()
4
5 for col in df.select_dtypes(include='object').columns:
6     df[col] = le.fit_transform(df[col])
7

```

### Problem 3: Feature Scaling

Fault Identified:

Features have different scales (e.g., Age: 18-25, Anxiety Score: 0-20). This variance can negatively affect distance-based algorithms like KNN and gradient-based algorithms like Neural Networks.

Solution Applied:

MinMaxScaler was applied to normalize all features to a range between 0 and 1.

```

10 scaler = MinMaxScaler()
11 X_scaled = scaler.fit_transform(X)

```

### Problem 4: Target Variable Selection

Solution Applied:

The target variables "Depression Value" (numerical score) and "Depression Label" (categorical severity) were removed from the feature set to prevent data leakage:

```

6 X = df.drop(['Depression Label', 'Depression Value'], axis=1)
7 y = df['Depression Label']

```

This ensures the model learns to predict depression levels based on symptoms and demographics rather than using the depression score itself.

## 4. Dataset Splitting

| Split        | Percentage | Method Used                      |
|--------------|------------|----------------------------------|
| Training Set | 80%        | train_test_split with stratify=y |
| Test Set     | 20%        | random_state=42                  |

Justification:

We used Stratified Sampling to ensure that both the training and testing sets have the exact same proportion of "Severe" vs "Mild" depression cases, preventing bias.

## 5. Model Training & Testing

### 5.1 Supervised Learning Models

#### Model 1: K-Nearest Neighbors (KNN)

Configuration: Number of neighbors (K) = 5.

Description: Classifies a student based on similarity to their 5 nearest neighbors.

#### Model 2: Decision Tree Classifier

Configuration: Default parameters (random\_state=42).

Description: Learns interpretable decision rules (e.g., "If Stress > 8, then High Risk").

#### Model 3: Logistic Regression

Configuration: Max iterations = 2000.

Description: A statistical baseline model used to estimate class probabilities.

#### Model 4: Neural Network (MLPClassifier)

Configuration:

- Hidden Layers: (64, 32)
- Activation: ReLU

- Max Iterations: 1000

Description: A Deep Learning model capable of capturing complex, non-linear relationships between academic pressure and mental health.

### 5.2 Unsupervised Learning: *K-Means Clustering*

We also treated the problem as unsupervised by ignoring the labels and applying K-Means Clustering (K=3) to find natural groups.

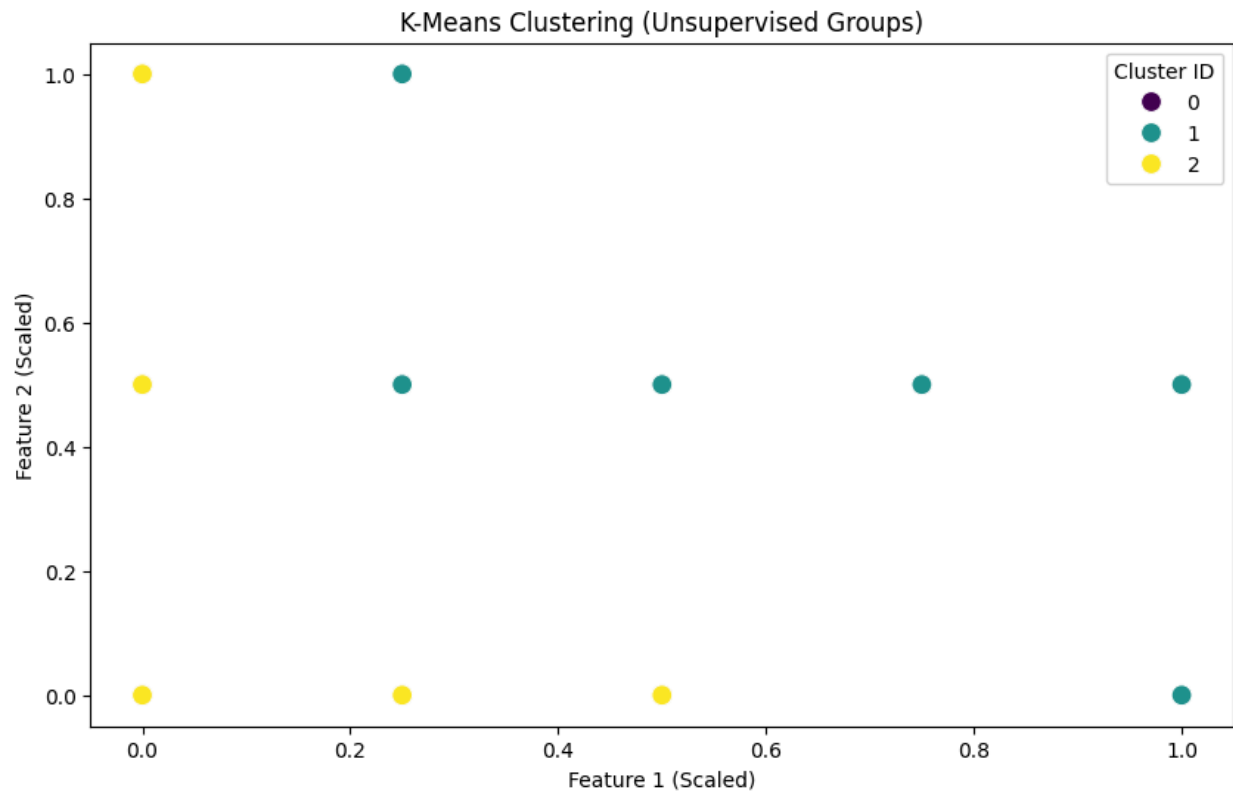


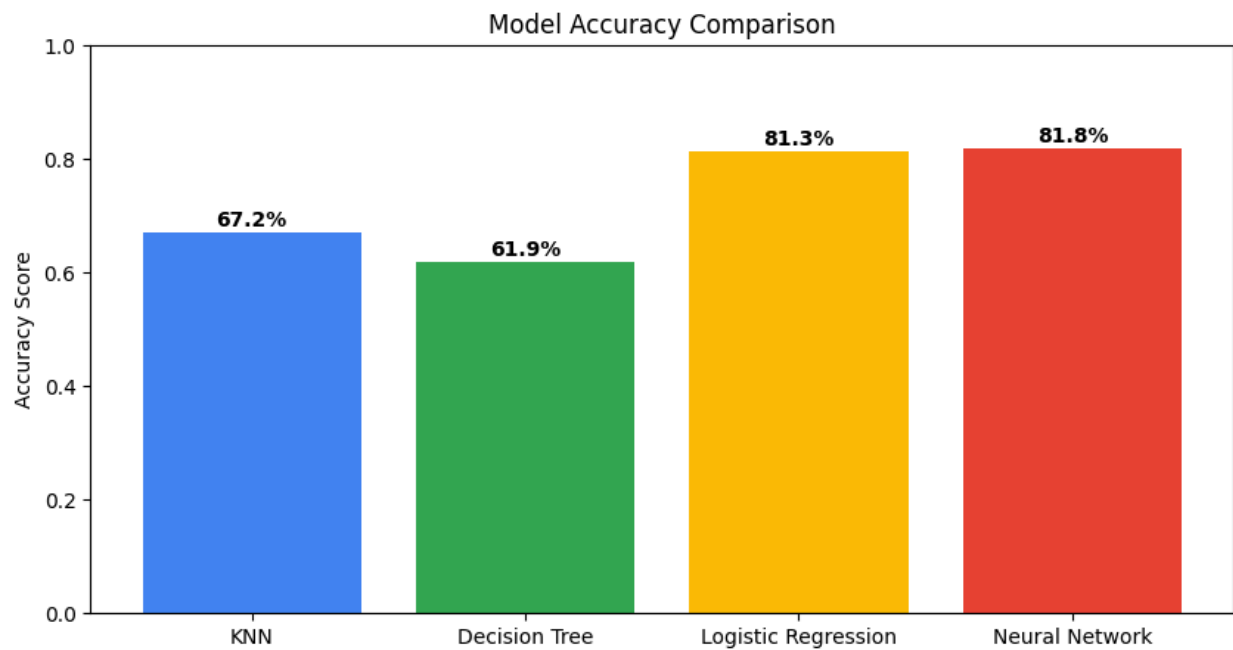
Figure 3: *K-Means Clustering showing 3 natural groups of students.*

## 6. Model Selection & Comparison Analysis

We evaluated all four models using the Testing Set.

### 6.1 Accuracy Comparison

The bar chart below compares the overall accuracy of each model.



*Figure 4: Accuracy comparison of KNN, Decision Tree, Logistic Regression, and Neural Network.*

## 6.2 Confusion Matrix Analysis

To understand where the models made mistakes, we plotted confusion matrices.

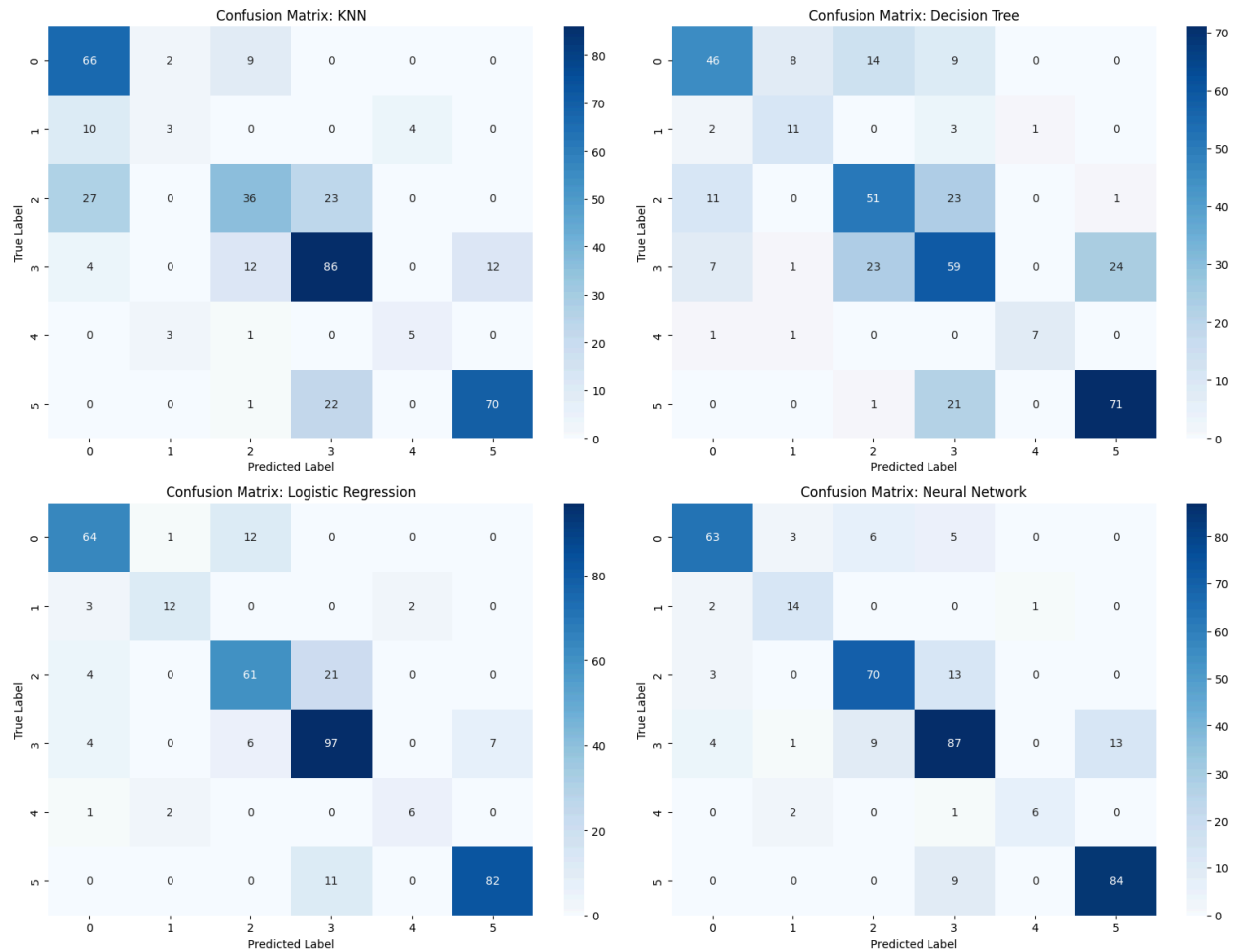


Figure 5: Confusion Matrices for all four models.

### 6.3 ROC Curve (Neural Network)

We analyzed the Receiver Operating Characteristic (ROC) curve for our Neural Network to check its ability to distinguish "Severe Depression" (Class 0) from other classes.

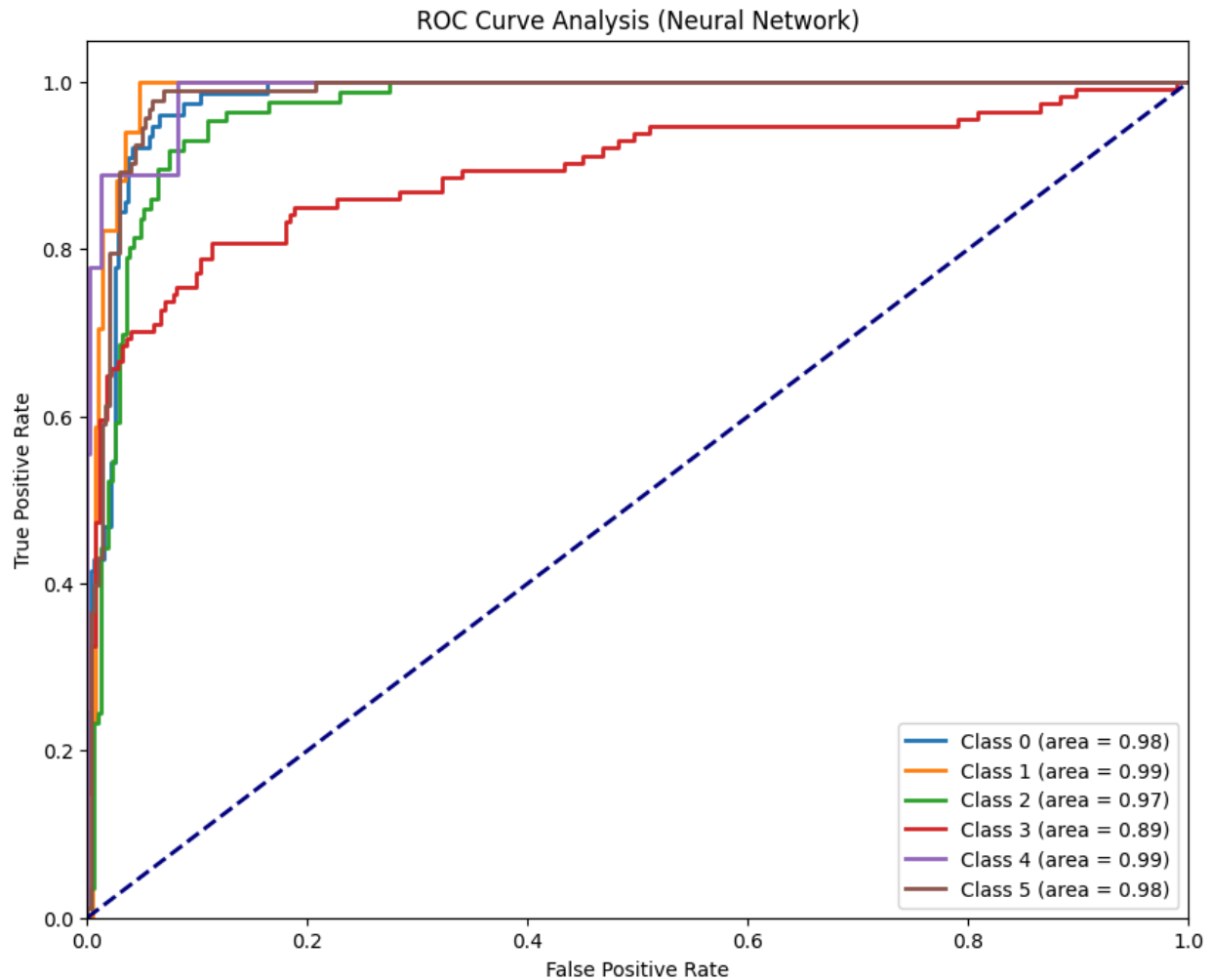


Figure 6: ROC Curve for the Neural Network model.

### 6.4 Comparison Summary

- Best Performer: The Neural Network achieved the highest accuracy(81.8%). It successfully captured the complex interactions between stress, sleep, and grades.
- Runner-Up: Logistic Regression also performed very well(81.3%), almost similar to Neural Network, but showed signs of slight overfitting compared to the Neural Network.

- Poorest Performance: Decision tree only predicted 61.9% which is the lowest of all four methods.
- Baseline: Logistic Regression provided a solid baseline, but struggled with the non-linear complexity of the data.

## **7. Conclusion**

### Findings:

Our results demonstrate that AI models can accurately predict student depression levels based on academic data. The strong correlation found between "Academic Stress" and "Depression" confirms that university workload is a major factor in mental health.

### Challenges Faced:

- Class Imbalance: The dataset had unequal numbers of "Severe" vs. "Mild" cases, which initially biased the models. We solved this using Stratified Splitting.
- Missing Data: Real-world data is rarely perfect. Implementing robust Imputation strategies was necessary to prevent the model from crashing.

### Future Work:

In the future, we could improve this system by collecting more diverse data (e.g., from different universities) and implementing a Real-time Web App where students can check their risk levels anonymously.